

Final Prototype Overview

The final prototype integrates the custom handle mechanism developed across the mockup and works-like phases onto a standard automatic umbrella platform with a custom handle casing. In addition to combining the lower-shaft string actuation and dual-lock system validated in earlier phases, the final prototype introduces three new elements. First, a ratchet-and-pawl mechanism housed within the custom handle casing enables incremental closure, so the user is never required to generate a single continuous pulling motion to fully collapse the canopy. Second, a 2:1 pulley system routed across the shaft reduces the force the user applies at the handle by half relative to the raw runner force. Third, a coordinated actuation link between the umbrella's existing release button and the ratchet pawl allows the user to disengage the pawl and reset the system in a single button press, without any secondary interaction.



Figure 1: Final Liminal Umbrella Prototype

The value-added custom components are the handle casing, which integrates the ratchet cavity, cord routing, and button linkage geometry into a single body; the ratchet-and-pawl assembly with a die-cast zinc internal mechanism and galvanized iron engagement surface; and the braided nylon pull cord terminating at the pull ring. Integrating onto an existing automatic umbrella platform rather than building a custom frame from scratch eliminates a substantial portion of the component count and manufacturing complexity, concentrating the custom value-add entirely in the handle where the user interaction differentiation lives. This is consistent with the broader lean manufacturing strategy of introducing custom integration points into an existing OEM production line rather than redesigning the full product architecture. Looking toward production, the handle casing is a clear candidate for transition from 3D-printed PLA to injection-molded ABS or nylon, which would reduce unit cost, improve surface finish, and enable tighter tolerances on the ratchet engagement geometry.

Overall, the final prototype successfully addresses the subspace of eliminating upper-shaft reach during closure. Lower-handle string actuation, incremental ratchet-controlled collapse, pulley-assisted force reduction, and coordinated button release are all validated within a handle envelope that integrates onto a standard umbrella platform. An additional planned development is routing the pull cord internally through the umbrella shaft rather than along the exterior, which would eliminate exposed mechanisms, improve the visual cleanliness of the product, and produce a more robust and protected cord path for everyday use.

The final prototype meets the reach specification at zero inches and the pull force specification at 3.8 kg (approximately 37.3 N), within the marginal target of 40 N. The incremental ratchet-controlled closure and reduced handle force together produce a coordinated and fluid closure motion, which aligns closely with what users described as their ideal interaction throughout testing. The team looks forward to continuing to develop the handle actuation system to a production-ready level.

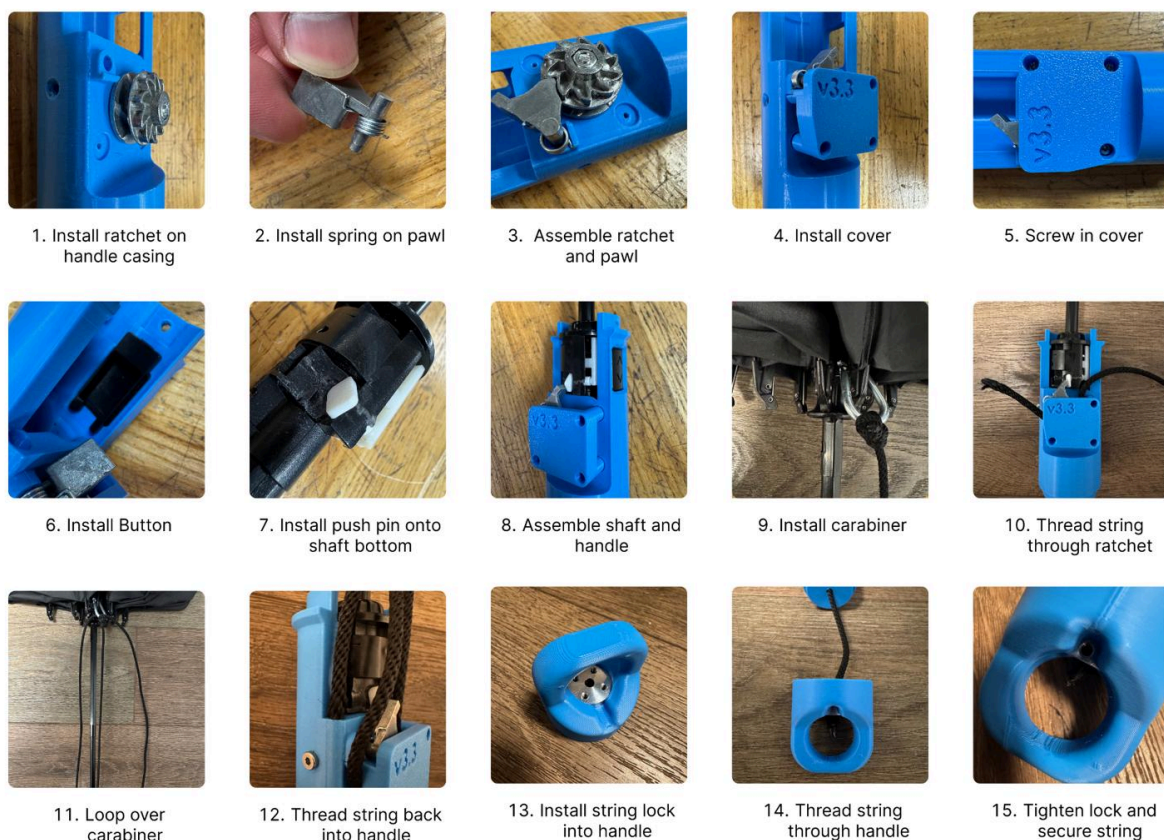


Figure 2: Final Liminal Umbrella SOP

A significant challenge emerged, however, around the integration of the container with the umbrella's actuation sequence. Collapsing the canopy into the container generates significant friction between the fabric and container walls, which can be reduced by increasing the diameter but at the cost of a larger overall form factor and a more cumbersome object to handle. Deploying the umbrella is even more difficult, as the automatic umbrella extends the shaft and deploys the ribs simultaneously, generating active friction against the container walls and preventing the umbrella from opening while seated in the container. This outcome is in part a reflection of the team's decision to concentrate prototyping resources on the handle actuation system, which user research consistently identified as the highest-friction moment in the umbrella experience. In hindsight, the problem is most cleanly understood as two independent subproblems: eliminating upper-shaft reach through handle actuation, and enabling integrated moisture containment through a compatible container architecture. The final prototype demonstrates that the first subproblem is well addressed. The second remains an open design space, and the team's view is that tackling it independently is the right path forward, with the long-term goal of converging both solutions into a fully integrated close-and-contain system.